Final Project Design Phase

Computer Science - Week 11

Oct 3, 2026 - Version 0.0.4

For complete instructions and project ideas, refer to the assignment in Canvas. Specifically, review the final project requirements before you continue below. If you have not reviewed the project requirements in Canvas (you might have to go to the modules view and check the current or past week's content), you will not be able to properly complete the parts below. In addition, it may be helpful to review the information on this Drafter documentation page.

0) Project Planning and Preparation

Successful programmers spend time planning before they begin coding. *After* reviewing the project requirements, answer the following questions to help guide your design and development process.

A. What concepts from class or previous assignments might help you complete this project? Which programming concepts (loops, conditionals, lists, classes, etc.) do you expect to use? What programming skills do you want to improve while completing this project?

I expect to use loops, conditionals and lists as a main part of my program. I want to get better at using Drafter.

B. Which parts of the project do you expect to be the easiest for you? The hardest? Why?

Everything else besides the Drafter will be easy for me, hopefully. I think making the database is going to be super annoying as well.

C. Even though you have not yet designed the final project, estimate which tasks will likely require the most time and effort? Which features or components should be implemented first?

The main page of the Drafter site and all of the pages. I'll try to inplement the helper functions first then the drafter site.

D. What distractions might interfere with your work and how will you minimize them? What resources will you need available while working? What questions, if any, do you still have about the assignment requirements?

All of my other final projects and exams that are due for other classes are important for me to work on. I have a research paper for my Honors Colloquium that requires some of my time. I'll try to dedicate my time towards this project the most since it's going to be hard to work on.

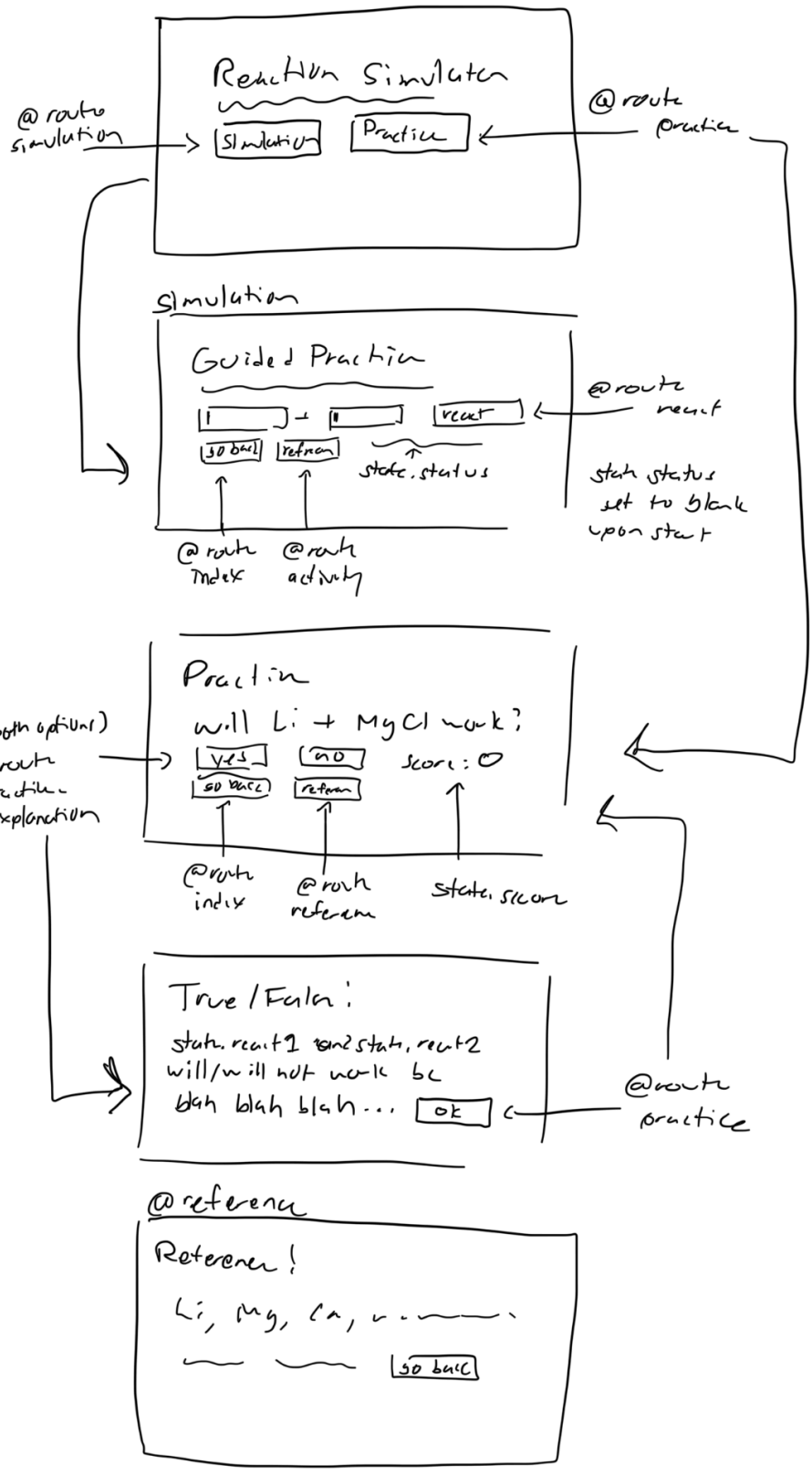
1) Site Diagram

Create a mock-up of your planned website. We suggest you make the mockup on paper, although you can also use software like PowerPoint or Slides to make a mockup. There is other design software that could also be used, but we strongly encourage you to keep it simple. Do not spend more than 15-30 minutes detailing the mock-up. Your goal should be to make a very quick and dirty diagram, to show that you have some kind of idea.

We recommend drawing each page as a rectangle, and laying out the text, buttons, and other components inside. Then, use arrows to indicate links between pages. Label each page the way you will name each function, and then put labels on the arrows to indicate what the button links. Also write down any parameters that should come into the pages.

In addition to the pages and routes, write down what fields your `State` will have, both their names and types. Think clearly about what data will need to be persistent across the entire website.

Add photo(s) or screenshot(s) of your website mock-up and labeled diagram on the next page.


Reaction Simulator
@route simulation
Simulation
Practice
@route practice
Simulation
Guided Practice
react
go back
refresh
state.status
@route react
state status set to blank upon start
@route index
@route activity
Practice
will Li + MgCl work?
(both optional) @route practice explanation
yes
no
score: 0
go back
refresh
@route index
@route reference
state.score
True / False:
state.react1 and state.react2 will/will not work bc blah blah blah...
ok
@route practice
@reference
Reference!
Li, Mg, Ca, ...
go back

2) Implementing Requirements

This table should serve as a draft for how you will implement the requirements into your final project. You may change your ideas later on, but your final site must fulfill all the requirements.

Requirement	**Implementation**
Purpose/Functionality A 1-2 sentence description of the purpose or functionality of your site	**Can combine two reactants together to form its predicted product. It will give a description of why it can or cannot work.**
State Dataclass Attributes Minimum of 4 attributes, one of which must be a list/dict of instances of your second dataclass. Briefly describe each attribute, its type, and how it will be meaningfully modified throughout the site.	class State: Reactant1: str - changes in textbox Reactant2 (with anion): str - changes in textbox Status: str - will change based on reactant1 and reactant2 and conditional statements. With given an error if reactant is not found in the database. Reaction : str - for selectionbox score : int - for practice mode. Will increase by one if user is right. answer : list[Answer] - for practice mode in the explanation page.
Second Dataclass Your second dataclass, including its attributes, their types, and how they will be meaningfully modified (if they are modified)	class Answer: Reactant1: str - chosen randomly in database for practice mode. Reactant2: str - chosen randomly in database for practice mode. will_work: bool - does manual calculation for why it won't work before reaction is shown to the user. why: str

	Will change based on reactants inputted. ex. “A reaction between (state.reactant1) & (state.reactant2) will (answer.will_work (react/not react). This is because (answer.why).
Routes (minimum 7) The names of your route functions	Index - Starting page where it asks to do simulation or practice mode Simulation_page Simulation_Check Practice_Mode Practice_Check Reference - shows activity series for oxidation Stop_practice - goes back to index react - takes in reactant1 and reactant2 and goes to simulation or practice check.
Pages (minimum 5) A brief description of the pages in your site	Index - Starting page where it asks to do simulation or practice mode Simulation - Takes you to do guided practice with reactant input. Simulation Explanation - state.status will change explaining why the reactants inputted do or do not work. Practice Mode - Can click will react or will not react button for practice given random reactants. Practice Explanation - after clicking the buttons previously listed, it will guide you to a page where it will explain why it will or won’t work. Reference - shows activity series for oxidation.
User Input (minimum 3) Description of user input components (TextBox, TextArea,	TextBox - For first reactant TextBox - For second reactant SelectBox - For either precipitate or aqueous reactions

CheckBox, or SelectBox) and their purposes For more information on these input types, see Drafter Documentation.	
If Statements (minimum 3) Brief description of what will be evaluated in your if statements and how they will meaningfully change the logic of the site	Simulation_check & Practice_check (they'll use the same format) - Compares the two reactants and then returns the result in the answer dataclass Score in Practice mode - if the right answer is chosen (either True or False) then add +1 to the score
Loop (minimum 1) This loop must iterate over the list/dict of dataclass instances in your State attribute. Description of this loop and why it is meaningful to the site	Assigns the reactants a number in the activity series– loops until the reactant is found in the database. (Line number indicates number)

LLM Conversation

If you used an LLM in any part of this work, copy and paste every part of that conversation below. Otherwise, you may be given a zero for the assignment. Obviously, these words will not all be yours, and this is the exception to the requirement at the top of this assignment that all responses be in your own words. If you did not use an LLM to assist you in this assignment, simply say so below.

Replace this text with your answer.

Final Submission

When you are finished, download this file as a Word Document (docx) and upload the file to the appropriate assignment on Canvas.